Lecture Scribe Example

L15_L17.pdf
PDF

I want a lecture scribe to refer for an exam. Here attached is the lecture notes for which I want you to give me exam ready scribe. The scribe should function as exam reference material and include proofs, derivations, and worked examples exactly as presented in the lecture.

Lecture Scribe Course:
Lecture number:
Topic:
Date:

Definitions:
(Key definitions)

Theorems / Propositions:
(Statements)

Proofs:
(Main proof ideas / steps)

Derivations / Equations:
(Important derivations)

Examples:
Given:
Find:
Solution:
Remarks:
Assumptions:
Special cases:
Summary:
Key results:

Above written is an example Scribe. Using the same

structure, generate a lecture scribe for the provided context.

Here is your **exam-ready lecture scribe** strictly structured from your provided notes (L15–L18). All derivations, steps, and examples are preserved exactly as in the lecture.

Lecture Scribe

Course: CSE400 – Fundamentals of Probability in Computing
Lecture number: L15–L18
Topic: Joint Random Variables & Joint Distributions
Date: March 10, 2026

Definitions

1. Joint Random Variables

Two (or more) random variables considered simultaneously to study their relationship.

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2. Joint PMF (Discrete Case)

If X and Y are discrete random variables:

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

- Represents probability of joint outcome occurring simultaneously
- Defined over product set:

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

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3. Joint PDF (Continuous Case)

A function $f(x, y)$ is a joint PDF if:

$$P((X, Y) \text{ in small area}) \approx f(x, y)\, dx\, dy$$

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4. Joint CDF

$$F(x, y) = P(X \leq x, Y \leq y)$$

Theorems / Propositions

1. Properties of Joint PMF

- $0 \leq p(x_i, y_j) \leq 1$
- $\sum_i \sum_j p(x_i, y_j) = 1$

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2. Properties of Joint PDF

- $f(x, y) \geq 0$
-

\int_c^d \int_a^b f(x,y),dx,dy = 1

]

- Note: $f(x, y)$ can be greater than 1 (density, not probability)

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3. Joint CDF Properties

- Non-decreasing in both variables
- Lower-left corner:

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$$

- Upper-right corner:

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

Proofs

1. Validity of Joint PDF (Normalization Proof)

Given:

$$f(x, y) = 4xy, \quad 0 \leq x \leq 1,\ 0 \leq y \leq 1$$

Check:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

Step-by-step:

$$= \int_0^1 \left[\int_0^1 4xy \, dx \right] dy$$

$$= \int_0^1 4y \left[\frac{x^2}{2} \right]_0^1 dy$$

$$= \int_0^1 4y \cdot \frac{1}{2} dy = \int_0^1 2y \, dy$$

$$= [y^2]_0^1 = 1$$

Hence:

- Total probability = 1
- $f(x, y) \geq 0$

✅ Valid joint PDF

Derivations / Equations

1. Joint PMF Table Representation

Organized as a matrix:

- Rows → values of X
- Columns → values of Y
- Entries → $p(x_i, y_j)$

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2. Continuous Case Relation

- Discrete → Continuous
- Sum → Double Integral

$$(X, Y) \in [a, b] \times [c, d]$$

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3. Joint CDF (Continuous Case)

$$F(x,y) = \int_c^y \int_a^x f(u,v)\,du\,dv$$

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4. Recovering Joint PDF

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

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5. Joint CDF (Discrete Case)

$$F(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

Interpretation:

- Sum all probabilities
- Rows: $x_i \le x$
- Columns: $y_j \le y$

Examples

Example 1: Rolling Two Dice

Given:

- Two fair dice
- X = value on first die
- T = total sum

Each outcome:

$$P(i,j) = \frac{1}{36}$$

Find:

Joint PMF of (X, T)

Solution:

Construct joint table (page 14):

- Entries = $1/36$ or 0
- Based on valid combinations

Remarks:

- Entries are either 0 or $1/36$
- X and T are **not independent**

Assumptions:

- Dice are fair
- Outcomes equally likely

Special Cases:

- Impossible sums → probability 0

Example 2: Continuous Joint PDF

Given:

$$f(x, y) = 4xy, \quad 0 \leq x \leq 1,\ 0 \leq y \leq 1$$

Event:

$$A = \{X < 0.5,\ Y > 0.5\}$$

Find:

1. Validity of PDF
2. $P(A)$

Solution:

Step 1: Validity

(Already proven above)

Step 2: Compute $P(A)$

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$$

Inner integral:

$$\int_{0.5}^{1} 4xy \, dy = 4x \left[\frac{y^2}{2} \right]_{0.5}^{1}$$

$$= 4x \cdot \frac{1 - 0.25}{2} = 2x(1 - 0.25)$$

$$= 2x(0.75) = \frac{3}{2}x$$

Now:

$$P(A) = \int_0^{0.5} \frac{3}{2}x \, dx = \frac{3}{2} \cdot \frac{x^2}{2} \Big|_0^{0.5}$$

$$= \frac{3}{4} \cdot (0.25) = \frac{3}{16}$$

Final Answer:

$$P(A) = \frac{3}{16}$$

Remarks:

- Region is upper-left rectangle in unit square (see diagram page 22)
- Uses geometric interpretation

Assumptions:

- Continuous uniform domain $[0,1]^2$

Special Cases:

- If region changes → limits change

Summary

- Joint distributions describe relationships between multiple variables
- Discrete → PMF (tables)
- Continuous → PDF (integrals)
- CDF connects probability and density
- Independence is not guaranteed
- Geometry plays key role in continuous case

Key Results

- $p(x_i, y_j) = P(X = x_i, Y = y_j)$
- $f(x,y)$ integrates to 1
- $F(x,y) = P(X \leq x, Y \leq y)$
- Continuous:

$$F(x,y) = \iint f(x,y)$$

- Discrete:

$$F(x,y) = \sum\sum p(x,y)$$

- Recovery:

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

If you want, I can also convert this into a **1-page cheat sheet**, **flashcards**, or **exam questions set** from this lecture.